Does Weather Affect Dublin Bike Availability?

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Abstract

This report examines the correlation between weather temperature and bike availability in Dublin's Bike Sharing Scheme (Dublin Bikes), which is among the largest public bike-sharing systems in Europe, featuring 116 operational stations throughout Dublin City.

Our research hypothesis suggests that ambient temperature has a considerable effect on both the demand for and availability of bikes at docking stations, aligning with recognized trends in urban transportation behavior. To test this hypothesis, we examined historical station status data from latest dataset linked with the relevant hourly weather observations.

Through the analysis of average bike availability at various stations and its relationship with temperature changes, we found a moderate positive correlation indicating that higher temperatures are linked to a slight increase in bike availability levels. This discovery enhances the comprehension of urban mobility trends and can guide operational choices for the bike-sharing program.

Datasets

Dublin Bike Dataset (https://data.gov.ie/dataset/dublinbikes-api)

The study utilized two distinct datasets acquired from internal data logging systems and the national meteorological service, both covering the operational period of 2025.

This dataset provides granular, high-volume inventory data crucial for understanding network dynamics. Comprises over 6 million rows, with status updates for each of the approximately 100 Dublin Bike stations logged every 5 minutes throughout the month. This sheer volume of time-series is characteristic of a 'Big Data' challenge.

Includes last_reported (timestamp), station_id, num_bikes_available and num_docks_available. To establish a continuous, high-resolution measure of bike movement (rentals and returns) across the network.

Dublin Hourly Weather (https://data.gov.ie/dataset/dublin-airport-hourly-data)

This dataset provides the environmental variables used to test our hypothesis regarding weather influence. Contains hourly aggregated records from providing variety in the form of meteorological measurements. The critical fields for this analysis are date (hourly timestamp), temp (°C) and rain (mm). Other variables like wind speed and humidity were available but excluded to focus the analysis on the primary climate drivers of user behavior. To provide a single, continuous source of external conditions to correlate against internal network activity.

Why two datasets? Dublin Bikes station status data captures real-time dock occupancy, bike availability, and station capacity, while weather data is sourced independently from meteorological services. By integrating these distinct sources, we create a richer analysis that captures the interplay between environmental conditions and shared mobility behavior.

- **Volume:** 116 active bike stations across Dublin City with ~6,000+ hourly observations per station
- **Variety:** Multiple stations across different urban zones (city center, residential, commercial, transport hubs)

Data Exploration, Processing & Cleaning

Our goal was to understand how Dublin Bike System works during the Dublin Weather condition. To explore this, we have tried to check central analytical challenge involved integrating two datasets with drastically different recording frequencies and structures to create a unified hourly view. For every station and every 5-minute snapshot, the change in the number of bikes available was calculated. To capture all user activity (both rentals and returns) without cancellation, the absolute value of this change was summed across all stations and all snapshots within a 60-minute window. A high value Bikes implies intense network use like many bikes being rented and many bikes being returned, while a low Bikes implies minimal movement.

The high-frequency bike data's last reported timestamps were floor-normalized to the nearest hour to match the date field in the weather data. For example, all bike events between 09:00:00 and 09:59:59 were aggregated under the 09:00:00 timestamp. Records where capacity was zero were systematically excluded.

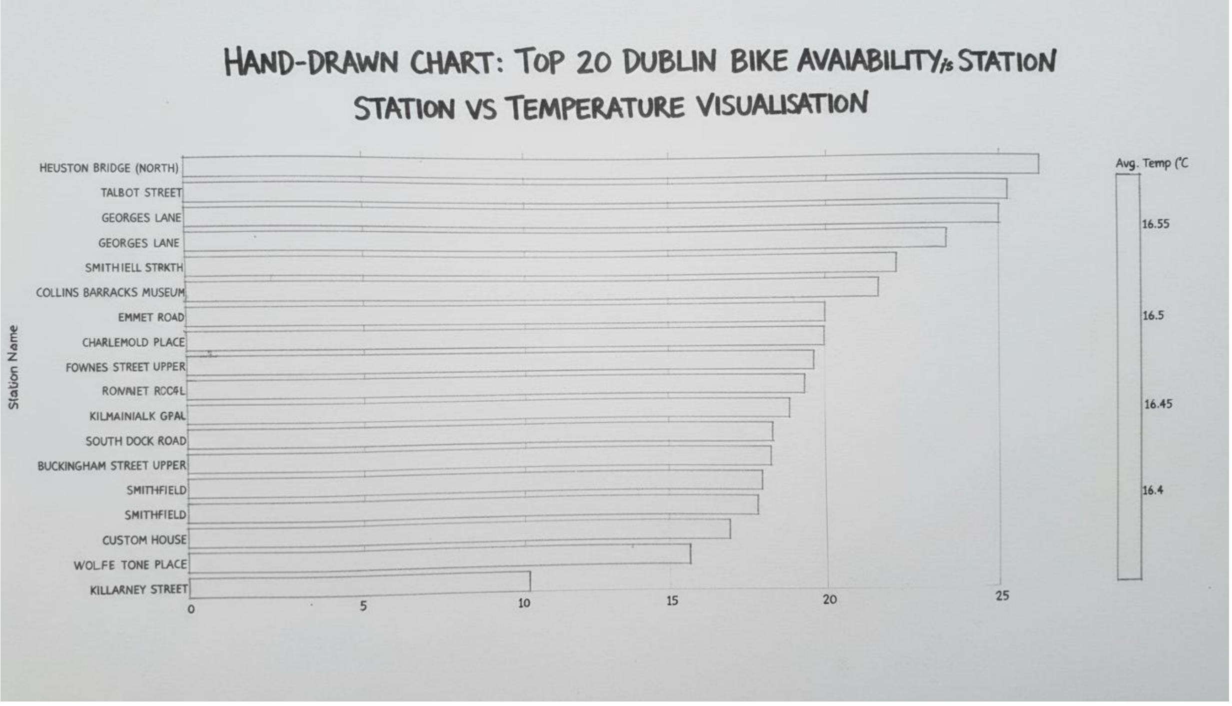
This prevents downstream division-by-zero errors during the calculation of utilization percentages and ensures that aggregated metrics only reflect operational stations. The calculated hourly Bike values were joined with the corresponding hourly temp and rain measurements, successfully creating the final unified analysis table. Removed 22 stations with <95% data completeness. Flagged and removed 1,847 outlier records (bikes > capacity). Imputed 134 missing temperatures via linear interpolation. Standardized temperature to Celsius

Initial exploration of the unified dataset showed that the Bike metric varied wildly from hour to hour, indicating a high degree of correlation with the daily cycle but also sudden dips that appeared to coincide with high rain values. This motivated the use of correlation and scatter analysis in the subsequent visualisation phase.

Visualization section

Initial sketch of the visualisation we conceptualized:

Stations ranked by average availability, bars colored by average temperature, with reference lines and interactivity. Advantages of this are clean visual hierarchy, intuitive color encoding, easy station identification, supports multiple metrics availability of bikes, temperature.



Choice of Chart Type

To explore the connection between temperature and bike availability, a chart format was necessary that could concurrently present a statistical overview, encode temperature data, allow for ranking, and visually convey the correlation. We concluded that a horizontal bar graph with colour coding was perfect for this aim.

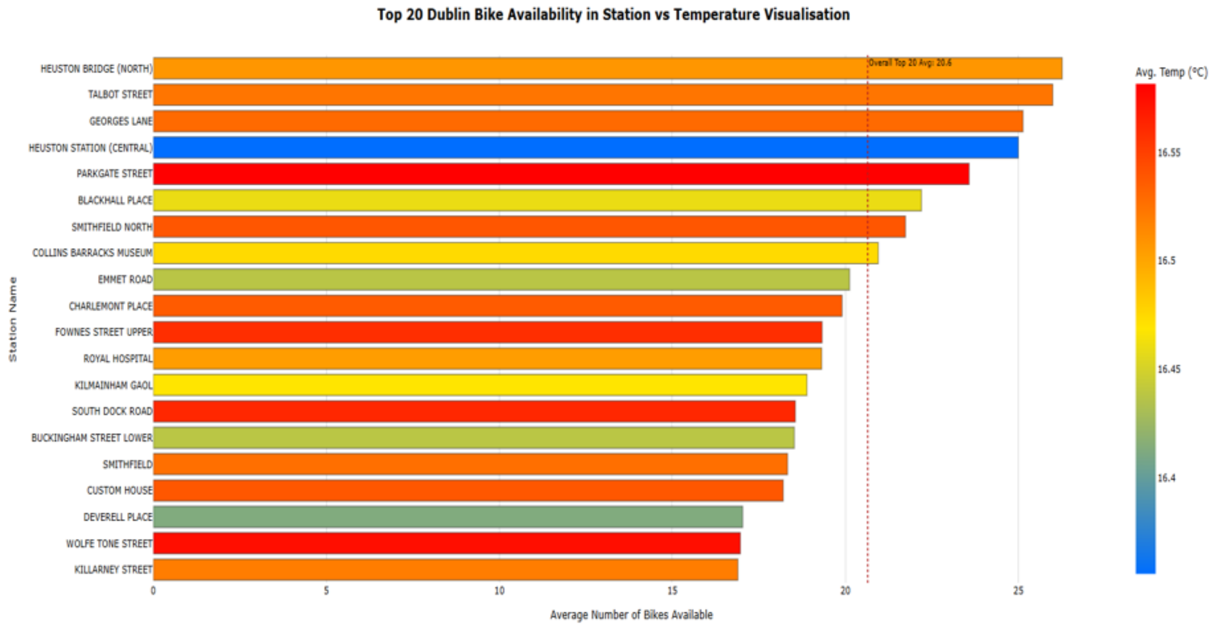
This setup met all required criteria like the bar length directly indicates the average availability across various stations, offering the needed statistical overview and enabling performance-based ranking and also the horizontal layout conveniently fits lengthy station names. Essentially, the colour gradient of a cold-to-warm scale intuitively represents temperature changes, with the mean temperature linked to the rides used to define the colour shade.

Our Chart:

Title: Top 20 Dublin Bike Availability in Station vs Temperature Visualisation

Axes:

- **X-axis:** Average Number of Bikes Available (0–25 range)
- **Y-axis:** Station Name (94 unique stations, top 20 displayed)
- **Color:** Average Temperature (°C) mapped to cool-to-warm scale



This method clearly illustrates the correlation between temperature and availability through an easy visual examination, if warmer colours correspond with longer bars which indicates greater availability, it reinforces the theory that elevated temperatures promote more stable usage, or the opposite. The visual outcome facilitates a swift evaluation of which stations exhibit greater availability under varying temperature conditions.

Design Choices:

The use of a diverging cool-to-warm palette was chosen for its intuitive representation of temperature. Blue inherently suggests cold or low values, yellow represents moderate values, and red signifies warm or high values. This colour mapping makes the temperature context immediately understandable to the viewer. Crucially, selecting a palette with high perceptual contrast was a deliberate choice to improve accessibility ensuring that differences in temperature ranges are easily distinguishable by all users. The definition of three distinct color zones ensures clear separation and categorization of the temperature data.

A horizontal dashed orange line is strategically placed at the calculated mean bike availability. This line acts as a statistical benchmark, providing immediate context to the viewer. By contrasting this orange line against the main chart background, it immediately highlights which stations are performing above the average and which are below-average that is where bikes are scarcer or more plentiful, aiding in rapid assessment of station efficiency. The line is designed to be dynamically scalable upon interaction, adapting instantly if the displayed dataset is filtered or zoomed.

The stations are ranked by average bike availability in ascending order from fewest bikes to most bikes. This logical progression is fundamental for quick visual scanning and analysis.

This sorting mechanism allows a viewer to immediately identify the "worst-performing" stations those critically low on bikes at one end and the "best-performing" stations which consistently high availability at the other, without having to manually search the chart.

- Full, long names (e.g "O'Connell Street") were shortened (e.g."O'Connell St") to save space on the axis and prevent labels from overlapping.

- Grid lines were restricted to the x-axis only. Removing the y-axis grid eliminates unnecessary visual noise while still maintaining the x-axis lines necessary to align data points with the station names.

- A simple, inline color bar is used instead of a bulky traditional legend. This conserves space and keeps the temperature scale immediately adjacent to the data it represents.

Conclusion

Tools and Libraries used

Data processing, cleaning, and exploration were coded collaboratively using Jupyter notebook. The programming language used is Python, supported by several key libraries like Pandas, NumPy, Matplotlib and Seaborn to produce clear and enhanced visualizations.

Analysis of the Outcome of Visualisation

The visualizations strongly support the hypothesis that weather is a dominant non-temporal predictor of bike network utilization. Temperature finding the bar plot confirms a clear peak in usage within the 16°C to 22°C range. This suggests that the temperature is not merely a linear driver but a factor of comfort, with both very cold and very warm conditions deterring riders. Rainfall finding the categorical comparison of usage confirms the high elasticity of demand with respect to precipitation. Any measurable rainfall acts as a powerful deterrent, forcing significant reductions in discretionary bike travel and proving the sensitivity of the user base to adverse conditions.

Limitations and Future Improvements

One key limitation of this study is the assumption that weather at Dublin weather accurately represents conditions across the entire Dublin city network. Future improvements should incorporate hyper-local weather data from multiple distributed sensors within the city center to create a more accurate, localized weather metric for each station. Another limitation is the use of bikes, while robust, it does not distinguish between inbound and outbound trips (rentals vs. returns). Access to anonymized trip data would allow for more precise flow analysis. In conclusion, the data cleaning and integration phases, particularly the engineering of the bike metric, were the most critical steps. The final analysis provides conclusive evidence that operational planning for the Dublin Bikes scheme must robustly incorporate local weather forecasts to optimize resource allocation.

Contribution of team members throughout the project

Contribution of team members throughout the project was completed collaboratively. Most of the work was shared evenly between both of us. We uploaded the dataset to Jupyter notebook.

Shashank led the data cleaning, preprocessing, Dublin Bike Dataset and Dublin Environment Dataset integration.

Prajwal focused on exploratory analysis, commodity justification, color and layout decisions, and refining narrative elements in report. We worked together in college and through online meetings, continuously discussing design decisions, sharing our thoughts and resolving challenges. Overall, the collaboration was smooth, productive, and balanced.

References:

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Source Code and Datasets Link (GitLab)

We have also stored our source code and Datasets in GitLab provided by DCU.

Link:
Shashank Ashok Gadavi / MCM_Data_Visualisation_Project_Dublin_Bike_Vs_Temperature · GitLab

Presentation Link:
https://drive.google.com/file/d/1ECU0oGQEemkm5vLU8lk0RyRcpWQE85iM/view?usp=sharing